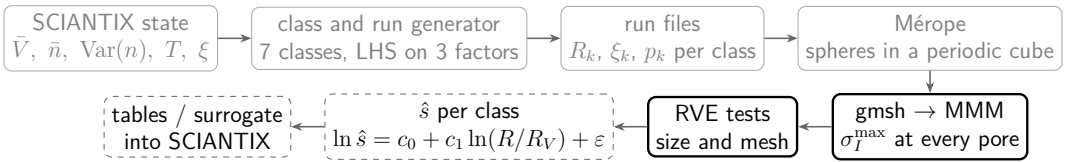
How Fission Gas Pressurised Pores Drive High Burnup Fuel Fragmentation: A Micromechanics-Informed Pore Population Model in SCIANTIX

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27 October 2026

Outline



- `bubble` case of mm-opera-hpc adapted to a population of pores: `polydispersebubble`
- pipeline Mérope → gmsh → MMM → analysis, first size and mesh tests on 10 cubes

Starting point: `bubble` (OperaHPC D5.2)

Elastic matrix Ω_m around pressurised pores, periodic cube:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \qquad \boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u}) \quad \text{in } \Omega_m$$

$$\boldsymbol{\sigma}\,\mathbf{n} = -p\,\mathbf{n} \ \text{ on the pores}, \qquad \mathbf{u} - \langle\boldsymbol{\varepsilon}\rangle\mathbf{x} \text{ periodic}, \quad \langle\boldsymbol{\varepsilon}\rangle = \mathbf{0}$$

$E = 150$ GPa, $\nu = 0.3$, lengths in μm.

$$p_{hyd} = -\tfrac{1}{3}\,\mathrm{tr}\Big(\tfrac{1}{V}\int_V \boldsymbol{\sigma}\,dV\Big).$$

Rupture and release

$$\sigma_I^{\max}\big|_i = \max_{|\mathbf{x}-\mathbf{c}_i|\le d_{min}} \sigma_I(\mathbf{x}), \qquad \mathrm{FGR} = \frac{\sum_i V_i \left[\sigma_I^{\max}\big|_i(p) > \sigma_R\right]}{\sum_i V_i}$$

From `bubble` to `polydispersebubble`

	`bubble`	`polydispersebubble`
pressure	one p for all the pores	one pressure per pore: $\boldsymbol{\sigma}\mathbf{n} = -p_{ref}\, p_i\, \mathbf{n}$ on pore i (6th column of the pore list)
search sphere	fixed $d_{min} = 0.95$ μm from the centre; pore of $R = 0.4$ μm: shell of 0.55 μm	$R_i + \delta$, here $\delta = 0.5$ μm: the same shell for every pore size
pores evaluated	near the global maximum ($\sigma_I > 0.9 \max$)	all the pores: the FGR needs $\sigma_I^{\max}\big\vert_i$ of every pore
output	points, `id x y z stress`	id, R, p, $\sigma_I^{\max}$, x, y, z; $\langle \sigma_h \rangle$, for every pore

$$\sigma_I^{\max}\big\vert_i = \max_{\mathbf{x}\in\Omega_m,\ |\mathbf{x}-\mathbf{c}_i|\leq R_i+\delta} \sigma_I(\mathbf{x})$$

Numerical and HPC choices

Solver

- Preconditioned conjugate gradient, the iterative solver of $K\mathbf{u} = \mathbf{f}$: BoomerAMG instead of the diagonal one. The diagonal one only rescales every equation: the iterations grow as the mesh is refined. AMG also corrects the error on coarser levels built from K: the iterations stay almost constant with the mesh size
- single-pore case: 100 * iterations with the diagonal one with respect to BoomerAMG;
- maximum number of iterations raised from 1000 to 5000 as a margin for the larger meshes

Memory and MPI

- every rank searches only its own maximum stress point for each bubble, then one gather of $[\sigma, x, y, z] \times n_{pores}$: exact global maximum per pore
- the original copied all the points on every rank: hundreds of millions with all the pores

Outputs

- pore list and $\langle \sigma_h \rangle$ always; ParaView fields on request

From the MMM run to the rupture criterion

Runs at unit pressure in every pore, zero macroscopic strain. Per pore $\sigma_i = \sigma_I^{\max}\big|_i / p$, and the mean hydrostatic stress of the matrix $\langle \sigma_h \rangle$ (< 0).

Back to zero macroscopic stress (superposition $\boldsymbol{\sigma}(p - p_{hyd}, 0) = \boldsymbol{\sigma}(p, p_{hyd}) + p_{hyd}\mathbf{I}$):

$$h = \frac{p_{hyd}}{p} = \xi - (1-\xi)\,\frac{\langle \sigma_h \rangle}{p}, \qquad \hat{s}_i = \frac{\sigma_i + h}{1-h}$$

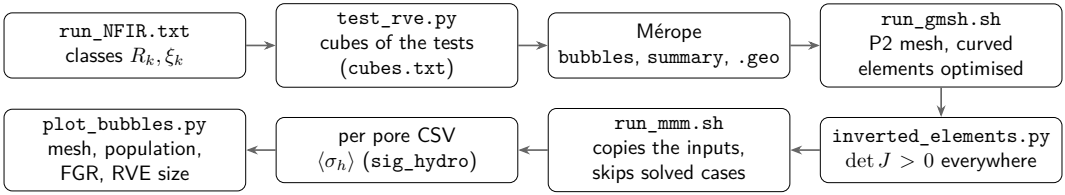
exact when all the pores have the same pressure.

Rupture and release: pore i breaks when $\hat{s}_i\, p > \sigma_R$, i.e. at $x_{c,i} = 1/\hat{s}_i$ with $x = p/\sigma_R$:

$$\mathrm{FGR}(x) = \frac{\sum_i V_i\, \mathbf{1}\big[\hat{s}_i\, x > 1\big]}{\sum_i V_i}$$

One MMM run gives the whole curve, for any σ_R.

Pipeline




```
test_rve_side10/run_NFIR/
  side10_seed01 ... side10_seed10      size test
  side10_seed01_meshdouble             mesh test
  side10_seed01_meshdouble_refinement  refined in MMM
  cubes.txt  unit_NFIR.txt
```

- one folder per cube: pore list, summary, mesh, MMM outputs
- the folder name tells the analysis what the cube is (side, seed, variant)

Data analysis

1. **Is the mesh fine enough?** Class means of $\hat{s}$ on a coarser mesh against the finest one, tolerance 2 %.
2. **Does a cube hold the whole population?** Pores expected and placed per class: a class with less than one pore per cube is missing from every cube, whatever the number of seeds.
3. **How much does the FGR change between seeds?** $x_c = p/\sigma_R$ at which every cube releases 10, 25, 50, 75, 90 % of the gas, compared between the 10 cubes.
4. **Which side gives the precision asked?** Kanit's method (next slide) on x_c and on the class means of $\hat{s}$.

Size of the RVE: the method of Kanit et al. 2003

Quantities, for one class of pores:

- Z: mean of $\hat{s}$ over the pores of the class in *one* cube (one value per cube)
- $\bar{Z}$: mean of Z over the $n = 10$ cubes (seeds)
- D^2: variance of Z between the cubes
- S^2: variance of $\hat{s}$ between single pores
- $V = L^3$: volume of the cube
- A: integral range, the volume over which $\hat{s}$ is correlated; for independent pores $A = 1/(\text{pore density})$ [HOW IS CALCULATED?]
- α: how fast the spread falls with the volume; $\alpha = 1$ for independent pores (variance of a mean of N values $= S^2/N$) [NON CAPISCO come lo calcoli]

Spread between cubes and error of one cube (95 %):

$$D^2(V) = S^2 \left(\frac{A}{V}\right)^{\alpha}, \qquad \varepsilon_{cube} = \frac{2D}{\bar{Z}}$$

One cube is an RVE if $\varepsilon_{cube} \le \varepsilon_R = 5\ \%$. Otherwise, with $\alpha = 1$:

$$n = \left(\frac{\varepsilon_{cube}}{\varepsilon_R}\right)^2 \text{ cubes of side } L,$$

or one cube of side

$$L_{RVE} = L \left(\frac{\varepsilon_{cube}}{\varepsilon_R}\right)^{2/3}$$

D^2 comes from only 10 cubes: its 95 % confidence interval (χ^2 distribution, 9 degrees of freedom) gives the range of L_{RVE} written next to it, e.g. 27 μm (21–40).

Check of $\alpha = 1$: independent pores give $A \times \text{density} = D^2 N_k / S^2 = 1$ (N_k pores of the class per cube).

Test set-up and cost

Pores: NFIR disc before the anneal, 7 classes $R = 0.22$–1.70 μm, $\xi = 0.114$, $R_V = 0.73$ μm; unit pressure in every pore; repulsion 0.3 μm; $\delta = 0.5$ μm.

Cubes: 10 cubes of 10 μm at mesh 0.3 μm (size test) + seed 1 at 0.6 μm (mesh test).

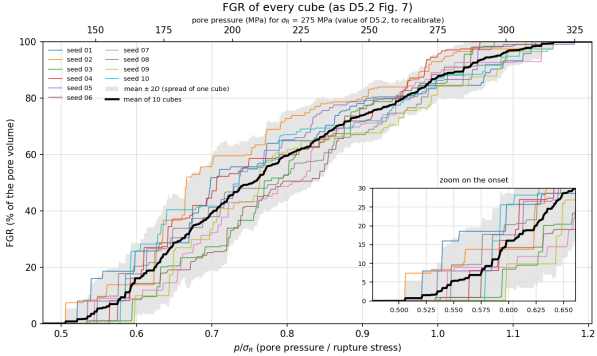
Planned: 30 μm at 0.15 μm, but on the cluster (should be available now).

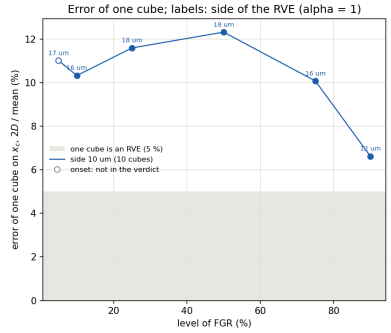
Does a 10 μm cube hold the population?

R (μm)	pore vol.	expected	placed
1.70	7.8 %	0.4	**0**
1.19	26.2 %	4.2	4
0.85	38.2 %	16.7	16
0.61	22.2 %	26.5	26
0.44	5.1 %	16.7	16
0.31	0.5 %	4.2	4
0.22	0.02 %	0.4	**0**

porosity 0.101 against 0.114: the 1.70 μm class needs a side ≥ 13.6 μm.

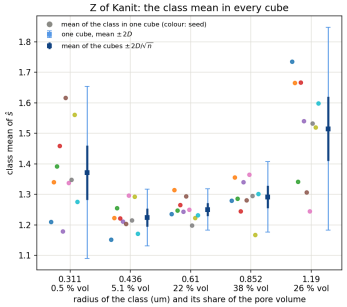
Result: FGR of the 10 cubes

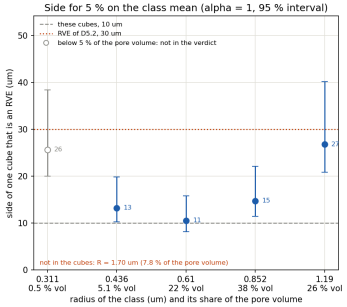


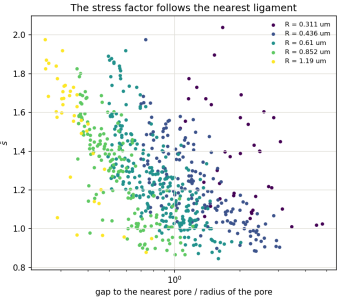


- $x_c = p/\sigma_R$ at which a cube releases 50 % of its gas: 0.747 ± 0.029 (mean of the 10 cubes, 95 %), i.e. $p \approx 205$ MPa with the $\sigma_R = 275$ MPa of OperaHPC (to recalibrate)
- error of one cube on x_c: 7–12 %, above the 5 % asked. On the FGR alone the RVE is ≈ 18 μm; 14–27 μm is its 95 % confidence interval (only 10 cubes)

Result: size of the RVE

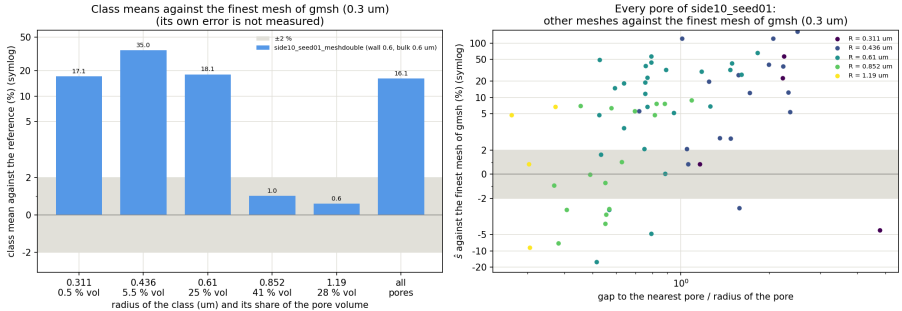






- **one 10 μm cube is not an RVE**: $L_{RVE} \approx 27$ μm (21–40), set by the 1.19 μm class (26 % of the pore volume), or the mean of 20 cubes of 10 μm; consistent with the 30 μm of OPERAHPC
- is the $1/V$ scaling right? If the pores of a class act independently, $A\times$density $= 1$. Measured 0.56–1.18, close to 1: extrapolating from 10 μm to L_{RVE} with $\alpha = 1$ is reasonable; a second cube side would measure α directly [TBD]
- $\hat{s}$ follows the nearest ligament: it grows when the gap to the nearest pore shrinks

Mesh convergence (seed 1, preliminary)



Same pores of seed 1 meshed at 0.6 and 0.3 μm. Left: class mean of $\hat{s}$ on 0.6 μm minus that on 0.3 μm, in % of the latter (grey band: ± 2 % tolerance). Right: the same pore by pore, against the gap to the nearest pore.

- the classes ≥ 0.85 μm (≈ 70 % of the pore volume) change by ≤ 1 %; the small ones by up to 35 %
- why it is not conclusive: with two meshes the reference (0.3 μm) is itself not converged.
- to redo with the mesh of the final runs: 0.15 μm

A pitfall of curved elements: inverted tetrahedra

A P2 tetrahedron is the image of a reference one through its 10 nodes; gmsh moves the mid-edge nodes onto the sphere so that the faces follow the pore:

$$\mathbf{x}(\boldsymbol{\xi}) = \sum_{a=1}^{10} \mathbf{x}_a N_a(\boldsymbol{\xi}), \qquad J = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}}$$

In a thin ligament between close pores (gap $\approx$ mesh size) the curved element folds over: $\det J < 0$ in part of it and the stiffness matrix may lose positive definiteness.

gmsh log: `Volume mesh:  worst distortion = -15.9 (..., 8 elements with jac.  < 0)`
MMM: `solve:  the non linear solver did not converge`

- 9 of 10 seeds (worst distortion -16 to -18): the linear solver fails
- seed 1 (worst -1.8) converged, but its 7 inverted elements all touch the 0.31 μm pores: $\sigma_I^{\max}/p \approx 1.65$ instead of ≈ 0.59 on three of them

Fix: gmsh `Mesh.HighOrderOptimize 2`.

Next steps and points to discuss

Next

1. mesh test at 0.15 μm on seed 1 (other PC, ≈ 40 GB)
2. a second side (20 or 30 μm) to measure α and the bias of the small cube; 30 μm on the cluster (`partition-mesh`: written, not yet tested in a full run)
3. fix side and mesh, then the LHS runs: $\ln \hat{s} = c_0 + c_1 \ln(R/R_V) + \varepsilon$ per run
4. tables / surrogate of $(c_0, c_1, \sigma_\varepsilon)$ into SCIANTIX

To discuss

- δ: 0.5 μm in these runs, larger than the 0.3 μm minimum gap: the search sphere can reach the ligament of the neighbour. $\delta \leq 0.3$ μm?
- **pressure per class** in the LHS runs: the rescaling to zero macroscopic stress is exact only with one pressure; otherwise a second run (unit isotropic strain, no pressure) and a linear combination
- **classes missing in small cubes** (1.70 μm, 7.8 % of the pore volume): the side follows the rarest large class